

Safety as a System Property From Control Design to Data-Driven Autonomy

Workshop Proposal for the 23rd IFAC World Congress

Organizers and Contact Information

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Motivation and Overview

Autonomous systems have advanced rapidly and are increasingly deployed in safety-critical domains, such as transportation, robotics, energy, and aerospace. As these systems operate with reduced human oversight in dynamic and uncertain environments and increasingly rely on learning-enabled components, failures can have severe consequences, making safety a primary requirement rather than a secondary concern.

Classical control theory provides powerful tools for stability, robustness, and safety in model-based settings, while recent advances in learning, optimization, and formal methods address uncertainty and complexity. In modern autonomous systems, however, tightly coupled perception, learning, planning, and control pipelines mean that safety cannot be fully understood by analyzing individual components in isolation. Instead, safety violations often emerge from interactions across layers and abstraction levels.

A key insight of this workshop is that safety must be treated as a *system-level property* spanning the full autonomy pipeline, from data and learning to decision making, planning, and control. This perspective raises fundamental questions on how heterogeneous safety guarantees (deterministic, probabilistic, distributional, and formal) can be composed and maintained under uncertainty, distribution shift, and system interaction effects. By bringing together ideas from robust and stochastic control, safety-critical control, distributional methods, system-level synthesis, and formal verification, the workshop aims to advance principled approaches for system-level safety assurance in data-driven autonomous systems. The topic is timely and directly aligned with the IFAC World Congress focus on systems and control.

Workshop Objectives and Topics

The primary objective of this workshop is to develop a unified perspective on safety as a system-level property in autonomous systems by examining how safety guarantees can be defined, compared, and composed across learning, planning, and control layers. The workshop brings together researchers from control theory, autonomy, learning, and verification to encourage cross-disciplinary exchange and to identify principled approaches for integrating heterogeneous safety frameworks in complex autonomous systems.

Topics of interest include, but are not limited to:

- System-level formulations of safety in autonomous systems operating under uncertainty;
- Safety-critical control design and analysis, including robust, stochastic, and distributionally robust methods;
- Integration of learning-enabled and data-driven components with formal or control-theoretic safety guarantees;
- Distributional and uncertainty-aware approaches to planning, estimation, and control;
- Safety in distributed, decentralized, networked, and multi-agent systems;
- Algorithmic and computational considerations, including scalability, robustness, and deployment;
- Case studies and applications in robotics, autonomous vehicles, energy systems, and cyber-physical systems.

Intended Audience and Expected Outcomes

The workshop targets researchers, practitioners, and graduate students interested in designing, analyzing, and assuring safety in autonomous systems under uncertainty. It is particularly relevant to those working at the intersection of control, optimization, machine learning, robotics, and cyber-physical systems, with a focus on safety-critical autonomy and system-level guarantees.

The workshop aims to (i) clarify and compare existing safety frameworks from a system-level perspective, (ii) identify key challenges in composing safety guarantees across learning, planning, and control, and (iii) outline emerging directions for scalable, deployable safety assurance. Through cross-disciplinary discussion, it seeks to bridge theoretical advances and practical autonomous systems.

Prerequisite Knowledge

The workshop is intended for participants with a background in systems and control theory. Familiarity with linear systems, feedback control, and basic optimization is recommended. Prior exposure to robust or stochastic control, formal methods, or learning-based approaches in control and robotics may be helpful, but is not required. The workshop will combine foundational concepts with recent developments, enabling participants with diverse backgrounds to engage with both classical and modern approaches to safety in autonomous systems.

Structure and Schedule

This *full-day* workshop, running from 9:00 AM to 5:30 PM, will feature nine invited talks, each consisting of a 40-minute presentation including a 5-minute Q&A. The workshop is organized into four technical sessions, each focused on a complementary aspect of system-level safety in autonomous systems.

- **09:00 AM–09:10 AM: Opening remarks**
- **09:10 AM–10:30 AM: Session 1**
 - 09:10–09:50: Naira Hovakimyan
 - 09:50–10:30: Lars Lindemann
- **10:30 AM–10:50 AM: Coffee break**
- **10:50 AM–12:50 PM: Session 2**
 - 10:50–11:30: Yongxin Chen
 - 11:30–12:10: Calin Belta
 - 12:10–12:50: Tigran Bakaryan
- **12:50 PM–02:20 PM: Lunch break**
- **02:20 PM–03:40 PM: Session 3**
 - 02:20–03:00: Gioele Zardini
 - 03:00–03:40: Karl H. Johansson
- **03:40 PM–04:00 PM: Coffee break**
- **04:00 PM–05:20 PM: Session 4**
 - 04:00–04:40: Angela P. Schoellig
 - 04:40–05:20: Navid Azizan
- **05:20 PM–05:30 PM: Closing remarks**

Speakers

The invited speakers collectively address the workshop’s central theme of safety as a *system-level property in autonomous systems*, bringing complementary expertise spanning learning-enabled and data-driven safety, robust and stochastic control, compositional and decentralized system design, multi-agent systems, and formal verification. Together, the talks span the autonomy pipeline and emphasize principled approaches to maintaining safety guarantees across interacting components and abstraction layers.

The speaker lineup reflects diversity across gender, career stage, and geography, with representation from institutions in North America, Europe, and Asia, and combines strong theoretical foundations with practical relevance. The tentative titles and abstracts for the confirmed talks are listed below; the full schedule will be available on the workshop website and included in the program upon acceptance.

Naira Hovakimyan (Director of the AVIATE Center, W. Grafton and Lillian B. Wilkins Professor of Mechanical Science and Engineering, University of Illinois Urbana–Champaign)

Ontological Robustness for Certification of Layered Autonomy Architectures

Abstract. Learning-based control paradigms have seen many success stories with autonomous systems in recent years. A typical architecture in these systems involves layers for perception, planning and control, wherein each of these layers uses different tools and metrics for assessing robustness and performance. For example, the planners – that use vision-based sensors to update the navigation and motion planning – operate largely relying on distributionally robust stochastic optimal control, whereas the low-level controller can be a deterministic controller with its conventional gain and phase (time-delay) margin. We present a new analysis framework for addressing this ontology challenge inherent to autonomous systems. We derive distributional robustness guarantees for deterministic L1 adaptive controllers that can be used by any stochastic planner without facing a language barrier. The combined planner-controller framework can serve as foundation for development of certificates for V&V of learning-enabled systems. An overview of different projects at our lab that build upon this framework will be demonstrated to show different applications.

Bio. Naira Hovakimyan received her MS degree in Applied Mathematics from Yerevan State University in Armenia. She got her Ph.D. in Physics and Mathematics from the Institute of Applied Mathematics of Russian Academy of Sciences in Moscow. She is currently W. Grafton and Lillian B. Wilkins Professor of Mechanical Science and Engineering and the Director of AVIATE Center of UIUC. She has co-authored two books, eleven patents and more than 500 refereed publications. She is the 2011 recipient of AIAA Mechanics and Control of Flight Award, the 2015 recipient of SWE Achievement Award, the 2017 recipient of IEEE CSS Award for Technical Excellence in Aerospace Controls, and the 2019 recipient of AIAA Pendray Aerospace Literature Award. In 2014 she was awarded the Humboldt prize for her lifetime achievements. In 2015 and 2023 she was awarded the UIUC Engineering Council Award for Excellence in Advising. In 2024 she was recognized as the winner of the College Award for Excellence in Translational Research, and in 2025 she was recognized for Excellence in Graduate Student Mentoring. She is Fellow of AIAA, IEEE, ASME, IFAC, and senior member of National Academy of Inventors. She has been named a Distinguished Lecturer for IEEE CSS for 2026-2028. She is a co-founder and chief scientist of Intelinair. Her work in robotics for elderly care was featured in the New York Times, on Fox TV, CNBC, and her recent NASA ULI award on flying cars led her to a live interview on Cheddar Innovates and many other media platforms. Her research interests are in control and optimization, autonomous systems, machine learning, neural networks, game theory, and their applications in aerospace, robotics, mechanical, agricultural, electrical, petroleum, biomedical engineering, and elderly care.

Calin Belta (Brendan Iribe Endowed Professor, University of Maryland)

Formal Methods for Safe and Interpretable Control

Abstract. In control theory, complicated dynamics such as systems of (nonlinear) differential equations are mostly controlled to achieve stability and to optimize a cost. In formal methods, simple systems such as finite state transition graphs modeling computer programs or digital circuits are analyzed or controlled from specifications such as safety, liveness, or richer requirements expressed as formulas of temporal logics. Many current applications, such as dexterous robotic manipulation, involve high-dimensional and partially known dynamical systems, which require machine learning techniques for motion planning and control. Ensuring safety and incorporating specifications given in rich, natural language within a learning-based system are challenging problems that received a lot of attention recently. I will show how techniques based on control barrier and Lyapunov functions can be combined with temporal logics and reinforcement learning to address these challenges. I will use examples from robotic manipulation and autonomous driving.

Bio. Calin Belta is the Brendan Iribe Endowed Professor of Electrical and Computer Engineering and Computer Science at the University of Maryland, College Park, where is also part of the Maryland Robotics Center (MRC) and the Institute for Systems Research (ISR). His research focuses on making control and machine learning systems safe and Interpretable, with particular emphasis on robotics and systems biology. Notable awards include the AFOSR YIP, the NSF CAREER award, and the HSCC Test of Time award. He is a Fellow of the IEEE.

Gioele Zardini (Rudge (1948) and Nancy Allen Career Development Assistant Professor, Massachusetts Institute of Technology)

Compositional Design of Society-Critical Systems: From Autonomy to Future Mobility

Abstract. When designing complex systems, it is necessary to consider trade-offs across multiple abstraction levels and scales, and to study component-level design choices jointly. In the context of future mobility, the design of autonomous vehicles and the mobility systems they enable are tightly coupled. This talk presents a framework for the compositional co-design of complex sociotechnical systems, leveraging monotone co-design theory and tools from game theory. The framework is instantiated through case studies spanning city-level policy design and autonomous mobility-on-demand services, demonstrating how the proposed approach enables modularity, interoperability, and scalable system-level optimization. Open challenges and future research directions in compositional system design are also discussed.

Bio. Gioele Zardini is the Rudge (1948) and Nancy Allen Assistant Professor at the Massachusetts Institute of Technology. He is a PI in the Laboratory for Information and Decision Systems (LIDS) and the Department of Civil and Environmental Engineering (CEE), and an affiliate faculty member of the Institute for Data, Systems and Society (IDSS). He received his Ph.D. in 2024 from ETH Zurich, where he also earned B.Sc. and M.Sc. degrees in Mechanical Engineering and Robotics, Systems and Control. Prior to joining MIT, he was a Postdoctoral Scholar at Stanford University and held visiting positions at nuTonomy Singapore (Aptiv/Motional), Stanford, and MIT. His research focuses on the compositional co-design of complex and autonomous systems, with applications ranging from future mobility to large-scale sociotechnical systems. He is the creator of Autonomy Talks and a lead organizer of the workshops “Compositional Robotics: Mathematics and Tools” and “Co-Design and Coordination of Future Mobility Systems” at IEEE ICRA and ITSC. His honors include the ETH Medal, the 2025 DARPA Young Faculty Award, Best Paper awards at ITSC and the Applied Category Theory Conference, and research awards from DoE, MIT, AFOSR, SIDARA, and Amazon.

Yongxin Chen (Associate Professor, Georgia Institute of Technology)

Safety Assurance of Stochastic Systems

Abstract. Safety is a critical requirement for real-world systems, including autonomous vehicles, robots, and power grids. While many methods exist for safety verification and control of deterministic systems, real-world applications also involve stochastic uncertainties that render such approaches insufficient. This talk presents an effective framework that addresses this challenge by decoupling the effects of stochastic and deterministic disturbances. A key component is a novel technique that provides high-confidence probabilistic bounds on deviations between stochastic system trajectories and their deterministic counterparts, applicable to both continuous- and discrete-time systems. These bounds reduce stochastic safety verification to a deterministic problem, enabling the use of existing methods to address uncertainty. The framework’s effectiveness is demonstrated through several safety verification and safe control examples.

Bio. Yongxin Chen was born in Ganzhou, Jiangxi, China. He received his BSc in Mechanical Engineering from Shanghai Jiao Tong University, China, in 2011, and a Ph.D. degree in Mechanical Engineering, under the supervision of Tryphon Georgiou, from University of Minnesota in 2016. He is currently an Associate Professor in the School of Aerospace Engineering at Georgia Institute of Technology. Before joining Georgia Tech, he had a one-year Research Fellowship in the Department of Medical Physics at Memorial Sloan Kettering Cancer Center with Allen Tannenbaum from 2016 to 2017 and was an Assistant Professor in the Department of Electrical and Computer Engineering at Iowa State University from 2017 to 2018. He received the George S. Axelby Best Paper Award (IEEE Transaction on Automatic Control) in 2017 for his joint work “Optimal steering of a linear stochastic system to a final probability distribution, Part I” with Tryphon Georgiou and Michele Pavon and the SIAM Journal on Control and Optimization Best Paper Award in 2023. He received the NSF CAREER Award in 2020, the Simons-Berkeley research fellowship in 2021, the A.V. ‘Bal’ Balakrishnan Award in 2021, and the Donald P. Eckman Award in 2022. He delivered plenary talks at the 2023 American Control Conference and the 2024 International Symposium on Mathematical Theory of Networks and Systems.

Lars Lindemann (Assistant Professor, ETH Zürich)

When Control Changes the Data: Safety under Interaction-Driven Distribution Shifts

Abstract. Accelerated by rapid advances in machine learning and AI, there has been tremendous success in the design of learning-enabled autonomous systems in areas such as autonomous driving and robotics. These exciting developments are accompanied by new fundamental challenges regarding the safety and reliability of these increasingly complex systems due to imperfect learning, system unknowns, and uncertain environments. Statistical tools for uncertainty quantification have gained popularity due to their ability to address these challenges. However, their guarantees rely on i.i.d. data, an assumption that is violated when control actions change the underlying data distribution. In this talk, new insights are provided for designing safe controllers under distribution shifts using robust conformal prediction (CP). The presentation begins by advocating for the use of CP due to its simplicity, generality, and efficiency compared to existing optimization-based verification techniques. It then introduces CP and summarizes existing work that uses CP to design probabilistically safe controllers in dynamic environments. Subsequently, interactive settings are considered in which the system's behavior may influence the environment's behavior, and vice versa. This circular dependency creates an interaction-driven distribution shift that invalidates existing CP guarantees. To address this problem, an iterative framework is proposed that episodically updates the controller while robustly maintaining safety guarantees by quantifying the potential impact of controller updates on the environment's behavior. This is achieved via adversarially robust CP, where a standard CP step is performed in each episode using observed data under the current controller, and safety guarantees are transferred across controller updates by analytically adjusting the CP results to account for distribution shifts. Finally, these ideas are extended to handle policy-induced distribution shifts that arise when using barrier or Lyapunov functions to control uncertain systems.

Bio. Lars Lindemann is currently an Assistant Professor for Algorithmic Systems Theory in the Automatic Control Laboratory at ETH Zürich. From 2023 to 2025 he was an Assistant Professor in the Thomas Lord Department of Computer Science at the University of Southern California. From 2020 to 2022 he was a Postdoctoral Fellow in the Department of Electrical and Systems Engineering at the University of Pennsylvania. He received his Ph.D. degree in Electrical Engineering from KTH Royal Institute of Technology in 2020. His research interests include systems and control theory, formal methods, machine learning, and autonomous systems. Professor Lindemann received the Outstanding Student Paper Award at the 58th IEEE Conference on Decision and Control and the Student Best Paper Award (as an advisor) at the 60th IEEE Conference on Decision and Control. He was finalist for the Best Paper Award (as an advisor) at the 2024 International Conference on Cyber-Physical Systems, the Best Paper Award at the 2022 Conference on Hybrid Systems: Computation and Control, and the Best Student Paper Award at the 2018 American Control Conference.

Tigran Bakaryan (Leading Researcher, Institute of Mathematics NAS RA, Center for Scientific Innovation and Education; Assistant Professor, Yerevan State University)

Fourier-Based Optimization for Constraint-Aware and Safety-Critical Optimal Control

Abstract. We consider continuous-time optimal control problems with state and control constraints and present a spectral optimization framework that is well-suited for safety-critical and system-level control applications. The approach represents state and control trajectories using Fourier series and enforces system dynamics through global norm-based constraints, enabling principled handling of hard constraints without relying on pointwise collocation. To support system-level safety analysis beyond idealized periodic settings, we extend Fourier-based methods to non-periodic problems using Eckhoff-type techniques that improve convergence and mitigate boundary effects. By exploiting the L_2 -orthogonality of the Fourier basis, the resulting formulation admits a compact quadratic structure, yielding reduced computational complexity and efficient numerical implementation. Numerical experiments demonstrate accuracy, robustness, and favorable performance compared to polynomial-based spectral methods, highlighting the potential of the approach as a scalable building block for constrained and safety-aware autonomous control systems.

Bio. Tigran Bakaryan is a Leading Researcher at the Center for Scientific Innovations and Education at the Institute of Mathematics of the National Academy of Sciences of Armenia, and an Assistant Professor at

Yerevan State University. Tigran received Bachelor's and Master's degrees from Yerevan State University (YSU). Earned Ph.D. in Fourier Analysis at the Institute of Mathematics of the National Academy of Sciences of Armenia. During Ph.D. studies, obtained a Young Researcher grant from the Science Committee of RA. Later, Tigran worked as a junior researcher at the Institute of Mathematics and concurrently as a teaching associate at the American University of Armenia (AUA) and YSU. In 2018, he joined King Abdullah University of Science and Technology (KAUST) as a postdoctoral fellow and worked on the theory of Mean-Field Games (MFG). During the post-doctoral program at KAUST, Tigran spent a few months at the Sapienza University of Rome and the University of Catania as a visiting research scientist. In 2023, he joined the Department of Mechanical Science and Engineering of the University of Illinois Urbana-Champaign (UIUC) as a postdoctoral research associate. At UIUC, he focused on applied problems related to multi-agent autonomous systems. His research interests are in the theory of PDEs, game theory, MFG theory, control theory, multi-agent autonomous systems, and related areas.

Navid Azizan (Alfred H. (1929) and Jean M. Hayes Assistant Professor, Massachusetts Institute of Technology)

Towards Safe and Efficient AI-Enabled Autonomy

Abstract. As AI models continue to deliver impressive performance, their deployment in real-world autonomous systems is becoming feasible. However, reliable deployment in safety-critical settings remains challenging due to unpredictable behavior under distribution shifts, limited ability to enforce hard constraints, and data inefficiency. This talk presents recent advances in improving the safety and reliability of learning-enabled autonomy stacks. First, a framework for incorporating hard constraints into neural network training via differentiable projection layers is introduced. Second, run-time monitoring methods based on uncertainty estimation and out-of-distribution detection are discussed to mitigate risks from unforeseen operating conditions. Finally, control-oriented meta-learning and continual learning approaches are presented to enable fast, data-efficient adaptation in dynamic environments. Together, these advances support safe and efficient data-driven control and decision-making with intrinsic safety and stability guarantees.

Bio. Navid Azizan is the Alfred H. (1929) and Jean M. Hayes Assistant Professor at the Massachusetts Institute of Technology, with dual appointments in the Department of Mechanical Engineering (Control, Instrumentation & Robotics) and the Institute for Data, Systems & Society (IDSS), and is a Principal Investigator in the Laboratory for Information & Decision Systems (LIDS). He previously held the Esther and Harold E. Edgerton (1927) Chair. His research spans machine learning, systems and control, mathematical optimization, and network science, with a focus on reliable intelligent systems and principled learning and optimization methods for autonomy and sociotechnical systems. He received his PhD in Computing and Mathematical Sciences from the California Institute of Technology in 2020, his MSc in Electrical Engineering from the University of Southern California in 2015, and his BSc in Electrical Engineering with a minor in Physics from Sharif University of Technology in 2013. Prior to joining MIT, he completed a postdoctoral appointment at Stanford University and was a research scientist intern at Google DeepMind. His work has been recognized with research and best paper awards from Google, Amazon, MathWorks, and IBM, and honors at venues including ACM Greenmetrics, L4DC, and INFORMS JFIG. He was named an Outstanding Academic Leader in Data by CDO Magazine in 2023 and 2024, received the 2020 ITA "Sun" (Gold) Graduation Award, and was named an Amazon Fellow in AI and a PIMCO Fellow in Data Science. His mentorship has been recognized with MIT's Frank E. Perkins Award for Excellence in Graduate Advising and the UROP Outstanding Mentor Award. He also co-founded and organized the "Control Meets Learning" Virtual Seminar Series.

Karl H. Johansson (Swedish Research Council Distinguished Professor, KTH Royal Institute of Technology)

Safe Learning and Control of Mixed-Autonomy Traffic

Abstract. Road infrastructure remains significantly underutilized in today's traffic systems. In mixed-autonomy environments, where connected and automated vehicles (CAVs) share the road with human-driven vehicles, there is substantial potential to improve efficiency, resilience, and sustainability. By leveraging CAVs

not only as transportation units but also as mobile sensors and distributed actuators, traffic systems can benefit from rich real-time data and coordinated control capabilities, enabling more accurate prediction and active management of traffic dynamics. This talk presents recent advances in physics-informed machine learning for modeling and control of mixed traffic. We demonstrate how hybrid modeling approaches that integrate traffic flow theory with data-driven learning lead to robust, scalable, and adaptive models capable of operating under dynamic and uncertain conditions. These models support proactive traffic control strategies that respond effectively to disturbances, reduce congestion, and enhance safety and environmental performance. We compare different machine learning architectures in terms of interpretability, scalability, data efficiency, and safety integration. We discuss how formal methods and uncertainty-aware reasoning can provide safety guarantees, and how teleoperation can serve as a structured fallback mechanism in safety-critical scenarios. The presentation will include extensive experimental results and real-world demonstrations developed in close collaboration with Swedish industry partners.

Bio. Karl H. Johansson is Swedish Research Council Distinguished Professor in Electrical Engineering and Computer Science at KTH Royal Institute of Technology in Sweden and Founding Director of Digital Futures. He earned his MSc degree in Electrical Engineering and PhD in Automatic Control from Lund University. He has held visiting positions at UC Berkeley, Caltech, NTU and other institutions. He is an Honorary Chair Professor of National Sun Yat-sen University. His research interests focus on networked control systems and cyber-physical systems with applications in transportation, energy, and automation networks. For his scientific contributions, he has received numerous best paper awards and various other distinctions from IEEE, IFAC, and other organizations. He has been awarded Distinguished Professor by the Swedish Research Council, Wallenberg Scholar by the Knut and Alice Wallenberg Foundation, Future Research Leader by the Swedish Foundation for Strategic Research. He has also received the triennial IFAC Young Author Prize, IEEE CSS Distinguished Lecturer, IFAC Outstanding Service Award, and IEEE CSS Hendrik W. Bode Lecture Prize. His extensive service to the academic community includes being President of the European Control Association, IEEE CSS Vice President, and Member of IEEE CSS Board of Governors and IFAC Council. He has served on the editorial boards of *Automatica*, *IEEE TAC*, *IEEE TCNS* and many other journals. He has also been a member of the Swedish Scientific Council for Natural Sciences and Engineering Sciences. He is Fellow of both the IEEE and the Royal Swedish Academy of Engineering Sciences.

Angela P. Schoellig (Alexander von Humboldt Professor of Robotics and Artificial Intelligence, Technical University of Munich)

TBD

Abstract. TBD

Bio. Angela Schoellig is an Alexander von Humboldt Professor of Robotics and Artificial Intelligence (AI) at the Technical University of Munich. She is a member of the Board of Directors at the Munich Institute of Robotics and Machine Intelligence (MIRMI) and serves as the Coordinator of the Robotics Institute Germany (RIG). Before moving to Munich in 2022, she was an Associate Professor at the University of Toronto, Canada, and a Faculty Member of the Vector Institute for AI. Angela conducts research at the intersection of robotics, controls, and machine learning. Her goal is to enhance the performance, safety, and autonomy of robots by enabling them to learn from past experiments and from each other. Angela has held a Canada Research Chair (Tier 2) in Machine Learning for Robotics & Control and a Canada CIFAR Chair in AI. She has also been a principal investigator of the NSERC Canadian Robotics Network. Angela is the recipient of an NSERC Arthur B. McDonald Fellowship (2022), the Robotics: Science and Systems Early Career Spotlight Award (2019), a Sloan Research Fellowship (2017), and an Ontario Early Researcher Award (2017). She is a Curious Minds Award winner (2022), an MIT Technology Review's Innovators Under 35 (2017), a Canada Science Leadership Program Fellow (2014), and one of Robohub's "25 women in robotics you need to know about" (2013). Her team is the four-time winner of the North American SAE AutoDrive Challenge (2018-21). Angela's PhD at ETH Zurich (2013) was awarded the ETH Medal and the Dimitris N. Chorafas Foundation Award. She holds an M.Sc. in Engineering Cybernetics from the University of Stuttgart (2008) and an M.Sc. in Engineering Science and Mechanics from the Georgia Institute of Technology (2007).

Equipment Requirements

No additional equipment beyond a standard projector, screen, and audio system (microphone and speakers) is required.